

# Bangladesh University of Business and Technology



**Course title:** Software Engineering lab

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**WBS report**

**ElectroFix: Used Electronic Buy, Sell and Repair Platform**

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# Software Requirements Specification

## 1 Introduction

### 1.1 Purpose

There are some purpose of our project. These are:

- **It helps people make money:** You can sell your old phone, laptop, or tablet that you don't use anymore. Instead of it collecting dust, you get cash for it.
- **It helps people save money:** You can buy a good-quality used device for a much lower price than a brand-new one. This is great for students or anyone on a budget.
- **It helps people fix things:** Instead of throwing away a broken device and buying a new one, you can get it repaired for less money. This saves you from a big expense.
- **It is good for the planet:** By repairing and reusing old devices, the project creates less electronic waste (e-waste). This means fewer broken devices end up in landfills, which is better for the environment.

### 1.2 Intended audience

#### People Who Want to Sell:

- Someone with an old phone sitting in a drawer.
- Someone who just got a new gadget and wants cash for their old one.

#### People Who Want to Buy:

- A student who needs a cheap laptop for school.
- Someone who wants a good device but doesn't want to pay a high price for a new one.
- A person looking for a temporary or backup phone.

#### People Who Need Repair:

- Anyone with a broken screen, bad battery, or other problem.
- Someone who wants to fix their device instead of throwing it away

### 1.3 Intended use

#### Objectives of the Platform:

- Enable users to list and sell their used electronic devices with detailed product descriptions, images, and pricing.
- Allow buyers to browse and purchase verified used devices through search and filter tools.
- Provide a streamlined repair service where customers can submit repair requests, get estimates, and track the repair process.

- Facilitate communication between users (e.g., buyer-seller, customer-technician).
- Ensure secure transactions through trusted payment gateways and verified logistics partners.
- Maintain platform quality through admin moderation, reviews, and reporting mechanisms.

#### **Target Audience:**

- **Individual Sellers:** Regular users or small resellers who want to earn money by selling used devices.
- **Buyers:** People seeking affordable, pre-owned devices from trusted sources.
- **Customers Seeking Repair:** Users with broken or malfunctioning devices looking for a fast and cost-effective repair service.
- **Repair Technicians:** Authorized partners or freelance service providers handling repair jobs.
- **Platform Administrators:** Users managing backend operations, verification, moderation, and support.

### **1.4 Product scope**

#### **This system enables:**

- Users to **sell** their used devices by creating listings.
- Buyers to **search, browse, and purchase** used devices.
- Customers to **request repairs**, track progress, and complete payments.
- Admins and partners to **manage inventory, repairs**, and user activities.

#### **The system will include:**

- User account management
- Product listing and search features
- Repair service module
- Payment gateway integration
- Admin panel for platform management

### **1.4 Risk definition**

This document outlines the potential risks associated with the buying, selling, and repairing of used electronic devices. These risks can impact system operations, customer satisfaction, legal compliance, and business continuity.

- **Data leakage from used devices:** Customer data may remain on devices during resale.
- **Device condition misrepresentation:** Devices may be inaccurately described during sale.
- **Stolen devices:** Some devices might come out as stolen leading to legal issues.
- **Repair Quality Issues:** Faulty or incomplete repairs can lead to customer dissatisfaction and returns.
- **Fraud:** Customer or third party may fraud during transactions.
- **Cyber threats:** Data breaches of customer information.
- **Inventory Loss or Theft:** Physical loss or theft of inventory due to weak controls.
- **Negative Reviews:** Poor service or products leading to bad reviews from customers.

## **2 Overall description**

### **2.1 User classes and characteristics**

#### **1 Buyer**

Characteristics:

- General public, age range can be 18-50
- Middle-class, lower-class family person
- Young people, students, old people, retired persons
- Can access the website through mobile
- Interested in buying an affordable device or cost consuming one

#### **2 Seller**

- Knowledgeable about the tech world at a medium to high level
- Can run a small business of tech products in a certain area
- Any general person wants to sell their device
- Want to make more profit than market value

#### **3 Technician**

- Skilled electronic device repair worker
- Can be a shop owner
- High and updated tech literacy
- Proper time management ability
- Experienced and trustworthy

#### **4 Admin**

- System owner or moderator
- High tech literacy
- Maintain the system
- Flexible to update the system as requirements change
- Ensure transparency
- Can have proper leadership

### **2.2 User needs**

#### **1 Buyer**

- Easy registration and login
- User-friendly interface
- Secure enough to do a transaction
- Trustworthy enough to buy the item
- Ensuring against fraud should be done.
- Proper transparency
- Option to trade in old device for a discount

- Easy search and filtered option

## **2 Seller**

- Good and user-friendly interface
- Easy product listing
- Customized price suggestions
- Ability to edit information of the products
- Ensure to get payment after selling
- A complaint box for any inconvenience
- Notification of trade in request
- Ensuring transparency
- Communicate with the buyer through the system

## **3 Technician**

- Register with specialization and area location
- Edit profile
- Flexibility to accept or reject a repair request
- Good environment
- Proper independence to work simultaneously
- Enough time
- customer ratings and reviews
- Work experienced update via system

## **4 Admin**

- Manage and approve
- Accept or reject buyer, seller, technician based on condition
- Database monitoring
- System controlling
- Can generate a report for each case
- Transparent users
- Skilled and trustworthy team member to handle the system
- Updated tool to reconfigure the system
- High security mode

## **2.3 Operating environment**

The system will operate in the following environments:

### **1. Client Side (User Devices):**

#### **Platforms:**

- Web browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari.
- Mobile applications:
- Android (version 8.0 and above).

- iOS (version 13 and above).

#### **Minimum Device Requirements:**

- Desktop/Laptop: Windows 10/11, Dual-core processor, 4 GB RAM,
- stable internet connection.
- Mobile Devices:
- Android: 3 GB RAM, Quad-core processor.
- iOS: iPhone 8 or newer.

## **2. Server Side (Backend):**

- Operating System: Windows 10/11.
- Web Server / Application Runtime: Node.js.
- Database: MongoDB (NoSQL database for high scalability and flexibility).
- Hosting Environment: Local server setup on Windows, or deployment to cloud platforms (Azure, AWS, or Google Cloud).
- Real-Time Communication: WebSocket or Firebase for expert chat and instant notifications.

## **3. Network Requirements:**

- Stable internet connection (minimum 2 Mbps recommended for clients).
- Secure HTTPS protocol with SSL/TLS encryption for all communications.

## **4. External Interfaces:**

- Payment Gateway APIs (PayPal, Stripe, or local providers).
- Translation APIs for multi-language support.
- Email/SMS services for alerts and notifications.

## **5. Security Environment:**

- Authentication via JWT (JSON Web Token).
- Role-based access control (Customer, Seller, Expert, Admin).
- Encrypted storage of sensitive data (using MongoDB security practices).
- Regular security updates and patches for Windows OS and Node.js environment.

## **2.4 Constraints**

### **1. Platform Constraints:**

- The backend system is required to run on Windows 10/11 environment.
- Database management is restricted to MongoDB, so relational database features (like SQL joins) cannot be directly implemented.
- The system must be compatible with major browsers (Chrome, Firefox, Edge, Safari) and mobile platforms (Android 8.0+ and iOS 13+).

## **2. Technology Constraints:**

- The application must use Node.js as the primary runtime environment for backend development.
- Only NoSQL database design patterns are supported due to MongoDB.

## **3. Security Constraints:**

- All data in transit must be secured using HTTPS with SSL/TLS.
- Sensitive user information must be encrypted and comply with standard security practices.
- Role-based access control (Customer, Seller, Expert, Admin) must be enforced.

## **4. Performance Constraints:**

- The system should support at least 10,000 concurrent users without downtime.
- The response time for common actions (searching, listing, or opening chat) must be under 3 seconds in normal network conditions.

## **5. Regulatory / Compliance Constraints:**

- The system must comply with data protection and privacy regulations.
- Identity verification for sellers (for “Verified Seller Badge”) must follow legal and ethical standards in the operating region.

## **6. Development Constraints:**

- Only open-source or licensed libraries/tools may be used.
- Integration with third-party APIs (payments, translations, notifications) must not violate their usage policies.

## **2.5 Assumption**

### **1. User Assumptions:**

- Users have access to a stable internet connection.
- Customers and sellers own compatible devices that meet the minimum system requirements.
- Sellers will provide genuine information and valid documents during verification for the Verified Seller Badge.
- Customers are familiar with basic smartphone or web navigation.

### **2. System Assumptions:**

- The system will operate primarily on Windows 10/11 servers for backend processing.
- MongoDB will handle all database storage and retrieval operations.
- Multi-language support assumes availability of translation APIs or pre-defined translation datasets.

**3. Operational Assumptions:**

- Payment gateways (Bkash, Rocket, Nagad, SSL commerce) will remain available and functional.
- Email/SMS gateways used for notifications will work reliably.
- External APIs used for verification or translation will provide consistent uptime and accuracy.
- Regular security patches and updates will be applied to Windows OS, Node.js, and MongoDB.

**4. Dependencies:**

- Dependency on third-party services (payment gateways, SMS/Email providers).
- Continued support for browsers and mobile OS versions used by customers.
- Verification process depends on seller cooperation and admin approval workflow.

### 3 Requirements

#### 3.1 Functional requirements

Adding all the user stories:

- Device verification with AI
- Notification Alert
- Local pickup & Delivery scheduling
- Real time price estimation
- Trade in option
- ownership & history verification
- Secure payment
- verified seller badge
- Multi language support
- Expert Chat system Device Verification with AI

##### Device verification with AI

Card: As a: Customer I want: To verify the condition of my device using AI tools So that: I can get the reasonable price to repair my device or to sell it.

Confirmation:

Success: 1. The AI successfully identifies the the device model and its specs. 2. It successfully identifies the the external damages from the uploaded image. 3. It also identifies internal mismatches . 4. It notifies and gives a proper report to the user about their device which they uploaded to verify.

Failure: 1. The AI doesn't complete its verification because of the poor quality of the image. 2. Verification is taking longer than usual time. 3. Shows improper ans reducing users trust.

##### Notification Alert

Card: As a: Buyer or Seller I want: To get clear notification alerts from the system So that: I don't miss important updates about my order or meeting time.

Confirmation:

Success: • System sends alerts for order confirmation, payment, and updates. • Users get reminders 24 hours and 1 hour before meeting. • Delivery status (on the way or reached) is sent on time. • Buyer and seller both get alerts at the same time. • Alerts reduce missed meetings or late deliveries.

Failure: • Buyer or seller may not get the alert due to network issues. • Notifications may arrive late or not at all. • Users may ignore or miss alerts if too many come. • If alerts are not clear, users may get confused. • Too many alerts can annoy users and they may turn them off.

## **Local Pickup & Delivery Scheduling**

Card: As a: Buyer or Seller I want: To easily schedule a specific time for local pickup or delivery when completing a purchase So that: The exchange of the item is convenient, secure, and fits into both of our schedules.

Confirmation:

Success: • System will ask buyer and seller to choose meeting time. • Both will confirm the time. • System will save and show the time in order details. • System will send confirmation message. • System will send reminders before meeting.

Failure: • Buyer and seller may not agree on a time, and order may get canceled. • Calendar may not load or notifications may fail. • A user may confirm time but not show up. • If system is hard to use, users may arrange time outside the app.

## **Real Time price estimation**

As a: Seller I want: to receive a real-time price estimate So that: I can set a fair price.

Confirmation:

Success: • Seller will provide complete product details. • System will process the data and show estimate price in display. • Seller can choose or set their price. • System will accept the chosen price and save it to database

Failure: • If mandatory information are missing, system will reject the product. • If the condition of the device is too poor, system will give a message: "Device condition not selected". • If there is no match for the device, system will display: "Unable to generate estimate price, please enter manually."

## **Trade in Option**

Card: As a: Buyer I want: Trade in my old device while purchasing another So that: I can reduce the cost and conveniently upgrade

Confirmation: Success: • When the buyer will give the valid details of their old device, it will accepted. • Then system will provide estimated Trade in Value. • Then the final cost will show to the buyer in display. • Buyer will confirm and complete check-out. • System will update transaction history.

Failure: • If the buyers old device is too old or not in the system database, system can't generate estimate Trade in Value. • The system will display to the buyer, "Your device is not eligible for Trade in. You can continue purchase without Trade in".

## **Ownership and History Verification**

card: As a:Admin I need to verify and provide ownership and history details of my electronic device. So that buyers can trust my product and I can sell it faster and count the profit and sell.

confirmation:

Success: • Seller uploads product details (brand, model, serial number). • System checks database for ownership and past history. • System confirms the device is genuine and not reported stolen. • Verification result is shown to the seller. • If the history is verified that will show a message to the seller “verified”.

Failure: • System checks the database for ownership and past history. • System finds mismatched data or missing buyers records. • Verification fails and seller gets a message: “Ownership/History cannot be verified.” • Product listing is not marked as verified until corrected.

## **security payment**

card: As a:Buyer i need a secure payment method. so that i can safely purchase or repair my used electronic devices without any risk with safe payment method.

confirmation:

Success: • Buyer choose the product and select the payment method which feels secure for him/her. • The system will show the buyer about payment information. • System checks and confirms the payment. • Money is sent safely, and both buyer and seller get a message.

Failure: • If there is any kind of technical issue(Bkash server down/transaction problems) the payment method will not work. • If mandatory information are missing, system will not work. • System tries to process but the payment fails. • Buyer gets a message: “Payment failed, please try again or use another method.”

## **Verified Seller Badge**

Card: As a: Customer I want: Sellers to have a verified seller badge So that: I can easily identify trustworthy sellers and feel confident while purchasing used devices.

Confirmation:

Success: 1. Sellers who complete verification (ID, business proof, or other checks) will get a badge on their profile. 2. Customers can clearly see the badge while browsing listings or chatting with the seller. 3. Verified sellers get higher chances of making sales. 4. Buyers report increased confidence and fewer concerns about fraud.

Failure: 1. Sellers fail to provide valid documents or information, so verification is denied. 2. Technical glitches prevent the badge from showing correctly on verified profiles. 3. Fraudulent sellers might attempt to bypass verification, which could reduce customer trust if not controlled.

### **Multi Language Support**

Card: As a: Customer I want: The platform to support multiple languages So that: I can use the app/website in the language I'm most comfortable with.

Confirmation:

Success: 1. Customers can easily switch between languages from the website settings. 2. Translations are accurate and understandable. 3. The entire user interface (UI) updates instantly when a new language is selected. 4. Customer will be satisfied using their own language.

Failure: 1. Some sections of the app remain untranslated or show mixed languages. 2. Poor or incorrect translations confuse customers. 3. Limited language options frustrate users if their preferred language is missing.

### **Expert chat System**

Card: As a: Customer I want: To chat with an expert So that: I can get the real time advice about my device its repairing, selling or buying.

Confirmation:

Success: 1. Customer gets instant reply about any queries 2. The chat system accurately handles the questions from the customer. 3. Chats are saved to the customer so that they can access it anytime they want. 4. Users are benefited through this chat system

Failure: 1. The chat system is taking a long time to response. 2. The chat system is giving inaccurate answers which is reducing customers trust. 3. Chats are not saved for that reason customer has to ask the questions again.

## **3.2 Non-functional requirements**

### **Performance:**

- The system has to be fast and well organized.
- It should handle many customers at a time without any kind of slowdown.
- The search option and other details should be easy and well organized, so that anyone can understand.
- Search results (products) should load within three seconds.

### **Security:**

- The system will have a high-security password for the admins so that no one can break the system easily.

- The history and confidential data (customer detail, information) should be encrypted.
- All payment methods will be used through a secure payment gateway.
- Payment assurance messages like "payment verified" must be provided.
- The customer has to log in with their email.
- The database must be secure, where data is fragmented so that the customer can only see the data they are allowed to access.

**Facilities:**

- The system should be simple and easy so anyone can use it.
- It should have a multi-language system.
- The verified customers will get a discount.

**Reliability & Availability:**

- The system should be available 24×7 for customer satisfaction.
- The payment system must be very secure and reliable.